CHE 422 BIOCHEMICAL ENGINEERING

Lecture Hours: Friday 1:00 pm – 3;30 pm, EIB 230

Instructor: Ying Liu, Office: EIB RM152, Telephone: 312-996-8249, Email: liuying@uic.edu

Textbook and reading materials:
Textbooks

Gene and Cell Therapy – C. Coutelle & J. Themis

Journals
- Science and Science series
- ACS Nano
- Nature and Nature Biotechnology
- Molecular Therapy
- Soft Matter
- PNAS
- …

Regulatory
- FDA Gene Therapy Guidance Documents

The course is aimed at senior undergraduate and graduate students and is designed to move engineers and other non-biologists past the initial language and knowledge barriers of the Biochemical Engineering field. This course provides a comprehensive introduction to gene therapy, covering the biological foundations, vector technologies, delivery strategies, clinical applications, regulatory considerations, and emerging innovations. Students will gain both conceptual understanding and practical insight into how gene therapies are developed, tested, and translated into approved treatments. The engineering concepts and processes of reactor design, quantitative thinking (kinetics, mass transfer, and fluid dynamics.), separation, scale-up, manufacturing, and control are integrated with the lecture topics.

Outline

Module 1: Foundations of Gene Therapy

Week 1 – Introduction to Gene Therapy

- History and evolution of gene therapy
- Monogenic vs complex diseases
- In vivo vs ex vivo approaches

Week 2 – Molecular Basis of Genetic Disease

- Gene expression and regulation
- Mutations and disease mechanisms
- Target selection and therapeutic rationale

Module 2: Gene Delivery Technologies

Week 3 – Viral Vectors I: Retrovirus & Lentivirus

- Vector design
- Integration and safety considerations
- Clinical applications

Week 4 – Viral Vectors II: AAV & Adenovirus

- AAV serotypes and tropism
- Immunogenicity
- Approved AAV therapies

Week 5 - 7 – Non-Viral Delivery Systems

- Lipid nanoparticles
- Polymer-based delivery
- Physical methods (electroporation, microinjection)

Module 3: Gene Editing & Regulation

Week 8 – Gene Editing Technologies

- CRISPR-Cas systems
- Base editing and prime editing
- Off-target effects

Week 9 – Gene Regulation & RNA-Based Therapies

- Antisense oligonucleotides
- siRNA and miRNA
- mRNA therapeutics

Module 4: Clinical Translation

Week 10 – Preclinical Development

- Animal models
- Biodistribution and toxicology
- IND-enabling studies

Week 11 – Clinical Trial Design

- Phases I–III
- Endpoints and biomarkers
- Patient recruitment challenges

Week 12 – Manufacturing & Scale-Up

- GMP requirements
- Vector production

- Quality control and CMC

Module 6: Frontiers & Future Directions

Week 13-14 – Advanced Therapies

- CAR-T and gene-modified cell therapies
- In vivo gene editing
- Synthetic biology approaches

Week 15 – Student Presentations & Capstone

<u>Methods of evaluation</u>

Distribution of final grade

Class attendance and discussion: 20%

- Active class discussion
- Class attendance

Reading assignment and journal club presentation: 30%

Mid-term project: 20%

Final project presentation: 30%

<u>Course policies</u>

Course format: **The class is designed for in-person participation only. For remote access of the course, please discuss with Dr. Ying Liu.**

Academic honesty: As a student of University of Illinois at Chicago (UIC), you have agreed to abide by the University's academic honesty policy, "A Culture of Honesty," and the Student Honor Code. All academic work must meet the standards described in "A Culture of Honesty". Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Questions related to course assignments and the academic honesty policy should be directed to the instructor.

Disability services: Concerning disabled students, the University of Illinois at Chicago is committed to maintaining a barrier-free environment so that individuals with disabilities can fully access programs, courses, services, and activities at UIC. Students with disabilities who require accommodations for full access and participation in UIC Programs must be registered with the Disability Resource Center (DRC). Center information can be found at http://www.uic.edu/depts/oar/campus_policies/disability_notification.html

Appropriate classroom behavior: The purpose of this is to ensure the equal right of learning.

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.